Mei Stoesz-Gouthro

V01004629

Methods

The study area is situated in the Pacific Ocean, along the West coast of Vancouver Island, British Columbia (see Figure 1). The Pacific Northwest is a coastal marine ecosystem and home to the humpback whale. This region is also vital for fishing vessels, fisheries, exploratory offshore wells, and marine tourism.

Humpback whales are prevalent in nearly all of British Columbia's waters but tend to cluster West of the Pacific Rim National Park and migrate North, towards inlets in the Calvert Island area, as stated by Storlund et al. (2024). In the off season, humpbacks are spotted at the origin point (see Figure 1) and are commonly sighted at the end point, later in the season. While marine mammals are constantly travelling, this least-cost path analysis represents a small, yet significant, portion of a humpback whale's journey.

The Pacific Ocean is a dynamic area, with both ecosystems and infrastructure, in the same areas. Humpback whales must avoid marine vessel, most notably, fishing vessels (Storlund et al., 2024). Offshore exploratory wells and tourist locations must also be avoided during migration. Fishing vessels pose the greatest risk to humpback whales, followed by offshore wells, and tourism. Though tourism is most prevalent off the East coast of Vancouver Island, it is significant enough to include in a weighted cost surface raster. The following equation demonstrates the weighted analysis:

$$cost\ surface = (0.5\ \times fishing\ vessel\ traffic) + (0.3\ \times wells) + (0.2\ \times tourism)$$

Together, these datasets demonstrate areas migrating humpback whales are most likely to avoid.

The binary wells dataset was reclassified to have scale values of 1 and 8, where 1 represents the absence of wells and 8 represents the presence of wells. Fishing vessel traffic and tourism datasets were reclassified with natural breaks, to emphasize the natural groupings, with minimal variance in classes and maximized variance between classes. Refer to Tables 2-7.

The least cost path algorithm is used to find a path from the given start and end points, with the least cumulative cost, using the cost surface raster. In this study, the least cumulative cost represents the lowest risk path for the humpback whale migration, based on fishing vessel traffic, wells, and tourism. The accumulated cost is calculated from the raster pixel that overlaps with the origin point. The next raster on the journey to the destination is selected based on the lowest weighted cost from all of the origin neighbours. This continues, until the path with the lowest accumulated path is achieved.

Finer resolution raster grids give more precise values for datasets, allowing for more path options from the start to the end point. Whale migration would be more accurate when compared to fine resolution rasters, since they swim in a line, vector-like, trajectory. Though fine resolution is optimal for precise analysis, it can be difficult to find fine resolution datasets.

The calculated least-cost path length for this study is 527.88 kilometres long. A study done in 2008, by Rambeau, found that whales have been re-sighted up to 845 kilometres away. This provides evidence to conclude the least-cost path is a realistic distance for humpback whales to swim, in British Columbia waters.

Table 1. Cost Analysis Dataset Metadata

	Dataset	Data Type	Projection	Spatial Resolution	Dataset Value Ranges
1	Fishing Vessel Traffic	Polygon	WGS84	5000m	0-1156 hours
2	Offshore Wells	Polygon	WGS84	5000m	0-10 wells
3	Tourism	Polygon	WGS84	5000m	0-2065 tours
4	Whale Start	Point	NAD83 / BC Albers	—	—
5	Whale End	Point	NAD83 / BC Albers	—	—

Table 2. Wells Dataset Original Values

Wells Original Classification		
Minimum	**Maximum**	**Scale Value**
0	0	0
1	1	1

Table 3. Wells Dataset Reclassification Values

Wells Reclassification		
Minimum	**Maximum**	**Scale Value**
-1	1	1
1	2	8

Table 4. Tourism Dataset Original Values

Tourism Original Classification		
Minimum	**Maximum**	**Scale Value**
0	5	1
5	10	2
10	15	3
15	16	4

Table 5. Tourism Dataset Natural Breaks Reclassification Values

Tourism Reclassification		
Minimum	**Maximum**	**Scale Value**
-1	0	1
0	2	2
2	4	3
4	6	4
6	8	5
8	10	6
10	12	7
12	16	8

Table 6. Tourism Dataset Original Values

Fishing Vessel Traffic Original Classification		
Minimum	**Maximum**	**Scale Value**
0	200	1
200	400	2
400	600	3
600	800	4
800	1000	5
1000	1156	6

Table 7. Fishing Vessel Traffic Dataset Natural Breaks Reclassification Values

Fishing Vessel Traffic Reclassification		
Minimum	**Maximum**	**Scale Value**
-1	4	1
4	15	2
15	33	3
33	57	4
57	91	5
91	157	6
157	326	7
326	1156	8

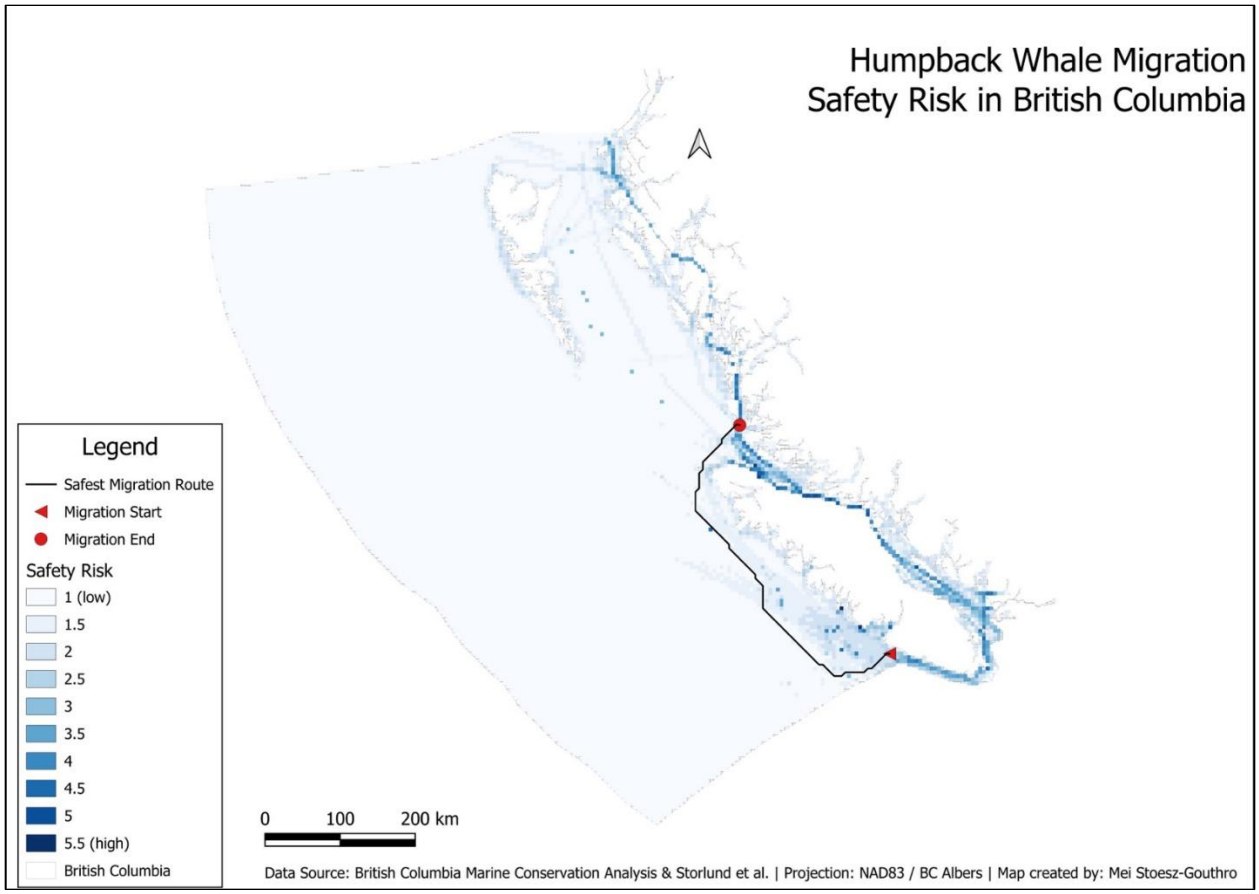

Figure 1. Least-cost path analysis of humpback whale migration in British Columbia. The cost surface was generated using weighted spatial layers representing fishing vessel traffic (highest weight), exploratory wells, and tourism hotspots (lowest weight).

References

Rambeau, A. L. (2008). *Determining abundance and stock structure for a widespread migratory animal : the case of humpback whales (Megaptera novaeangliae) in British Columbia, Canada* (T). University of British Columbia. Retrieved from https://open.library.ubc.ca/collections/ubctheses/24/items/1.0066809

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